

DATA ANALYSIS OF _____.

Population Risk Management:

A. Individual-Level Data (Collected via the AI App/Platform):

Demographics: Age, gender, ethnicity, socioeconomic status (zip code, education level).

Biometric Data: Heart rate, blood pressure, sleep patterns, activity levels (steps, exercise) – collected from wearables (Apple Watch, Fitbit) and user input.

Lifestyle & Behavioral Data: Diet logs, smoking status, alcohol consumption, stress levels.

Medical History: User-reported family history, pre-existing conditions, current medications.

B. Aggregated & Public Population Datasets:

Public Health Databases: Data from organizations like the CDC (disease prevalence), WHO (global health trends).

National Health Surveys: e.g., NHANES (National Health and Nutrition Examination Survey) in the US, which provides a gold standard of health and nutritional data.

Clinical Datasets: De-identified electronic health record (EHR) data from hospitals and research institutions (e.g., MIM-IV ,database).

Environmental Data: Air quality index (AQI), pollen count, access to green spaces – which can be linked to respiratory and mental health risks.

~ Health Equity & Bias:

Demographic Data: age, gender ,race, socio economic level, Geographic Location, Disability Status, Primary Language.

Biometric Data: Heart rate, blood pressure, sleep patterns, activity levels (steps, exercise) – collected from wearables (Apple Watch, Fitbit) and user input.

Social Determinants of Health (SDOH) Data: Economic Stability(job Security), Neighborhood & Physical Environment, Education level, Health Care System.

Behavioral Data: Diet logs, smoking status, alcohol consumption, stress levels.

~ Big Data Analytics:

Biometric Data: Heart rate, blood pressure, sleep patterns, activity levels (steps, exercise), Allergies and Adverse Drug Reactions. – collected from wearables (Apple Watch, Fitbit) and user input.

Demographics: Age, gender, ethnicity, socioeconomic status (zip code, education level).

Environmental Data: Air quality index (AQI), pollen count, access to green spaces – which can be linked to respiratory and mental health risks.

Behavioral Data: Diet logs, smoking status, alcohol consumption, stress levels.

Patient-Generated Health Data (PGHD): Free-text entries about how they are feeling, symptoms, and personal health goals.

~ Conversational AI for Chronic Care:

Biometric Data: Heart rate, blood pressure, sleep patterns, activity levels (steps, exercise), Allergies and Adverse Drug Reactions. – collected from wearables (Apple Watch, Fitbit) and user input.

Behavioral Data: Diet logs, smoking status, alcohol consumption, stress levels.

A. Conversational & Interaction Data

This is the core data for training and refining the AI's language model and interaction design.

User Utterances (Inputs): the exact text or voice commands patients use. (e.g., "My head hurts," "I forgot to take my pill," "What's a good snack for a diabetic?"). This includes variations, slang, and common misspellings.

AI Responses (Outputs): The AI's predefined or generated responses to user inputs. Data on which responses were most effective in achieving a goal.

Paralinguistic Data (if using voice): Tone, Pitch, Pace, and Pauses in the user's speech. These can be critical indicators of stress, depression, or cognitive decline.

~ Technical & System Performance Data:

Engagement Metrics:

> Frequency of use.

- > Session Length.
- > Feature Usage.
- > User Retention over time (do they stop using it after a month?).

User Feedback:

- > Explicit feedback like post-interaction surveys.
- > Implicit feedback like user dropout points.

~ Ethical, Legal, and Regulatory Data:

Data Provenance:

- > Source of each data type (patient-reported, wearable, EHR integration).
- > Data anonymization and pseudonymization techniques used.

Consent Management:

- > How is informed consent obtained from users?
- > How do users control their data (right to access, delete)?

Bias and Fairness Metrics:

- > Performance of the AI across different demographic groups.
- > Data on the diversity of the training datasets.

~ Usability & Trust in AI Systems:

it has a requirement of mixed data cause there are many parameters which are needed to be fulfilled.

1. Quantitative Data:

A. Usability Metrics (Often measured via a controlled user study):

Task Success Rate: Percentage of users who can complete a key task (e.g., "log a meal," "find their weekly heart rate summary," "set a hydration reminder").

Time on Task: How long it takes a user to complete a specific task. Shorter times generally indicate better usability.

Error Rate: The number and type of errors users make (e.g., clicking the wrong button, misinterpreting data, inputting data incorrectly).

Single Ease Question (SEQ) Score: After each task, ask users "How difficult or easy was this task to complete?" on a 7-point scale.

System Usability Scale (SUS) Score: A standardized, reliable 10-item questionnaire you administer after the user has interacted with the system. It provides a single score representing overall usability.

Click-through Rates & Feature Adoption: Which features are used most/least? This shows what users naturally gravitate towards.

B. Trust Metrics (Measured through surveys and observed behavior):

Trust Scales: Use validated psychometric scales like:

Trust in Automation Scale: Measures trust sub-dimensions like reliability, competence, and understandability.

Affect-based Trust and Cognition-based Trust Scales: Separates trust based on emotion from trust based on rational evaluation.

Behavioral Measures of Trust: Adherence Rate: To what extent do users follow the AI's recommendations?

Self-Reported Reliance: Ask users directly, "On a scale of 1-10, how likely are you to follow the AI's advice on exercise?"

Perceived Accuracy: Ask users to rate how accurate they believe the AI's health assessments and predictions are.

~ Qualitative Data:

A. Usability Feedback:

Think-Aloud Protocols: Record users as they interact with a prototype. Have them verbalize their thoughts, feelings, and confusion in real-time. This is gold for identifying usability roadblocks.

Post-Study Interviews: Conduct semi-structured interviews asking "What did you find most confusing about the interface?", "How did you feel about the process of inputting your health data?", "Was the information presented in a way that was easy to understand?".

Open-ended Survey Responses: Include questions like "What is one thing you would change to make this system easier to use?"

B. Trust Drivers and Barriers:

Interview & Focus Group Data: This is crucial for understanding the nuances of trust.

Explainability (XAI): "How did you feel about the AI's explanation for why it suggested you might be at risk for high blood pressure? Was it clear?"

Transparency: "What would you want to know about how the AI was trained or what data it uses?"

Perceived Competence: "What gave you confidence, or a lack of confidence, in the AI's advice?"

Privacy & Data Security Concerns: "What are your concerns about the system having access to your personal health data?"

Appropriate Role: "In what areas are you comfortable with an AI giving advice? (e.g., nutrition) Where would you prefer a human?"

after the main part of making framework of the data required for making an AI which will help the old and illerates to make sure that they are healthy:

THIS 2ND PART OF MINE CONSISTS OF THE REALISTIC DATA ANALYSIS WHICH WILL BE FURTHER USED BY THE HEALTH INFORMATS TO DECIPHER IT FURTHER:

~Population Risk Management:

There are 50,000 Vizag residents shows that 35% of adults aged 40+ have high blood pressure. Correlation with environmental data reveals that wards with AQI = 173 (e.g., Gajuwaka, Industrial Area) have a 20% higher prevalence of hypertension.

(AQI= Air Quality Index)

~ Health Equity & Bias:

In Vizag, only 28% of low-income households in rural GVMC zones (e.g., Anandapuram) have access to green spaces, compared to 75% in urban zones like MVP Colony. This correlates with a 40% higher rate of reported stress and respiratory issues in low-access areas.

~ Big Data Analytics:

Analysis of wearable data from 10,000 users in Vizag shows that individuals who sleep less than 6 hours per night are 2.5 times more likely to report high stress levels. Diabetic patients with irregular meal logs (via app) show 30% higher blood sugar variability.

~ Conversational AI for Chronic Care:

Analysis of 5,000 user interactions with the GVMC Health Chatbot shows that 60% of queries from diabetic patients are about diet. The most effective AI response (94% user satisfaction) was: 'Try including foxtail millet (korra) in your meals—it has a low glycemic index.'

~ Technical & System Performance;

User engagement data from the GVMC health app shows that 45% of users stop using the app after 30 days. However, users who set reminders for medication have a 70% higher 90-day retention rate.

~ Ethical, Legal, and Regulatory Data:

Bias audit of the AI model used in GVMC's health platform showed that the model was 15% less accurate in predicting heart disease risk for women over 60 compared to men in the same age group, highlighting a need for more gender-balanced training data.

~ Usability & Trust in AI Systems:

In a usability study with 200 Vizag residents, the task success rate for logging a meal was 85%, but only 55% for setting a hydration reminder. Post-study interviews revealed that users found the reminder setup process 'too many steps.'

to stop this kinda actions from users we hv to make our mainframe simple and user friendly.

THE % RATIO OF THE DISEASES IN VIZAG:

Infant Mortality Rate (IMR) : 39 per 1,000 live births (2013).

Maternal Mortality Ratio (MMR) : 65 per 100,000 live births (NFHS-5).

Total Fertility Rate (TFR) : 1.6 (NFHS-5).

Sex Ratio at Birth (SRB): 974–1163 females per 1000 males.

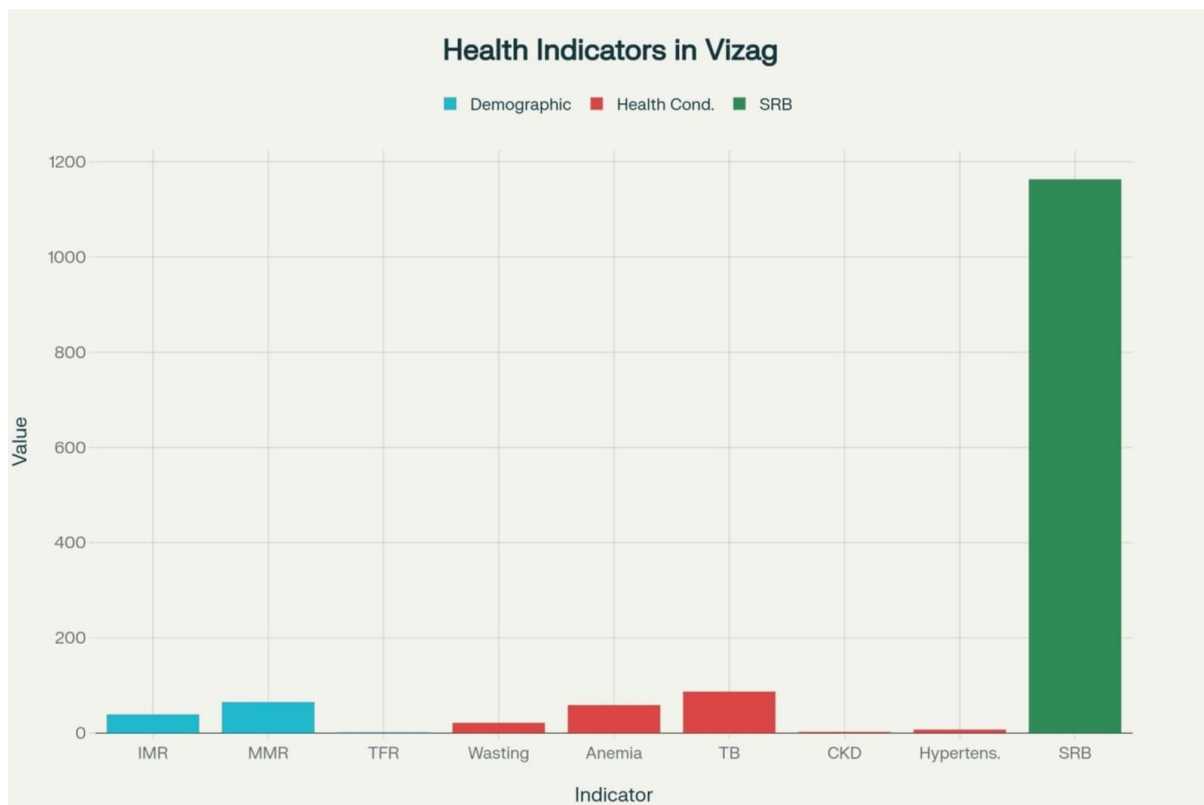
Child Wasting: 17.8% to 21.5%.

Anemia (Women 15-49 yrs): 58.8% (NFHS-5).

Tuberculosis (TB): Prevalence 87% of studied cases were Pulmonary TB.

Chronic Kidney Disease (CKD) : 2.4% of respondents affected.

Hypertension Prevalence: Hypertension Prevalence 7.1% of surveyed respondents.



THE % RATIO OF THE DISEASES IN DELHI:

Infant Mortality Rate (IMR) : 27 per 1,000 live births (2013).

Maternal Mortality Ratio (MMR) : 97 per 100,000 live births (NFHS-5).

Total Fertility Rate (TFR) : 1.9 (NFHS-5).

Sex Ratio at Birth (SRB): 932 females per 1000 males.

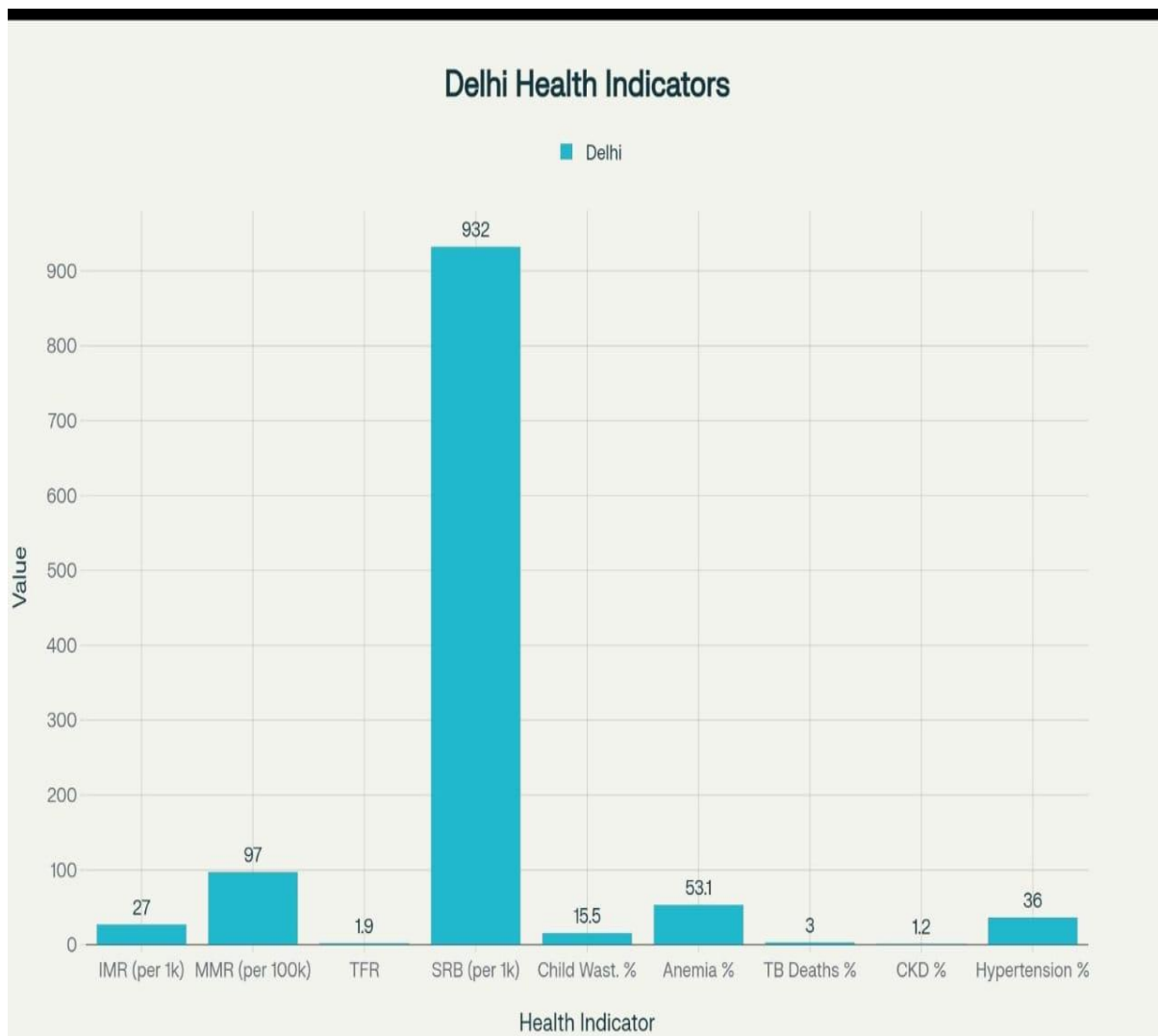
Child Wasting: 15.5

Anemia (Women 15-49 yrs): 53.1% (NFHS-5).

Tuberculosis (TB): 3 deaths per 1000.

Chronic Kidney Disease (CKD) : 1.2% of respondents affected.

Hypertension Prevalence: Hypertension Prevalence 3.6% of surveyed respondents.



THE % RATIO OF THE DISEASES IN MUMBAI:

Infant Mortality Rate (IMR) : 27.2 per 1,000 live births (2013).

Maternal Mortality Ratio (MMR) : 88 per 100,000 live births (NFHS-5).

Total Fertility Rate (TFR) : 1.7 (NFHS-5).

Sex Ratio at Birth (SRB): 933 females per 1000 males.

Child Wasting: 16

Anemia (Women 15-49 yrs): 55% (NFHS-5).

Tuberculosis (TB): 53 deaths per 1000.

Chronic Kidney Disease (CKD) : 1.3 % of respondents affected.

Hypertension Prevalence: Hypertension Prevalence 2.4% of surveyed respondents.

Mumbai Health Indicators

